models, it did not hold up well in the real world. However, researchers did not realize this during the early years of MVO.

What they did realize is that computers had limited power back in the 1950s, and MVO was computationally demanding. MVO could require matrix inversions involving covariances and returns of thousands of assets. A 1,000-asset portfolio, for example, would require 550,000 covariances.

Therefore, in the early to mid-1960s, a number of academic researchers working independently developed a simplified alternative to MVO called the capital asset pricing model (CAPM).[5]

CAPITAL ASSET PRICING MODEL

What early CAPM did was a linear regression between the excess return (return less the risk-free rate) of an asset (or portfolio of assets) and the excess return of the market index. A linear regression determines the relationship between two or more variables and provides measures that allow you to determine the accuracy of that relationship.

The beta coefficient of the CAPM regression equation tells you the sensitivity of an asset's excess return to variations in the market's excess return. In other words, it tells you how much the market's movement contributes to your return. CAPM also tells you that the expected return on any security is proportional to the risk of that security as measured by its beta.

The intercept of the regression equation is the alpha. It is what you have left over once you remove beta from the equation. Alpha represents abnormal profits. It is what you earn in excess of the reward for assuming market risk.

Instead of having to deal with possibly thousands of inputs to construct optimal portfolios, with CAPM you only need information pertaining to your portfolio of stocks and the market index. If you diversify among a number of stocks spread out in different industries, you could target the expected return and volatility you want for your portfolio by targeting the average beta of your stocks. Equally important from an academic point of view, you could use CAPM to help determine a firm's cost of capital and to